Automatic Barrier System

Homework 1

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1 Problem and Goal

The goal of this project was to create a small automatic barrier prototype in the Tinkercad Arduino environment. The system detects an approaching object using an ultrasonic distance sensor and automatically controls a servo motor representing a barrier.

In addition to moving the barrier, the system provides visual feedback using red and green LEDs and audible feedback using a piezo buzzer. The prototype therefore combines a sensor, actuator, visual indicators, and sound output into one automatic control system.

2 System Design

The HC-SR04 ultrasonic sensor continuously measures the distance to an object. When an object is detected at a distance of **150 cm or less**, the Arduino opens the barrier by moving the servo from 0° to 90°.

The barrier does not immediately close when the measured distance slightly changes. A separate closing threshold of **170 cm** is used. This creates hysteresis and prevents the barrier from repeatedly opening and closing when an object is close to the threshold.

If no valid ultrasonic echo is detected while the barrier is open, the system waits for **2 seconds** before closing it.

When the barrier opens:

- the green LED turns on;
- the red LED turns off;
- the buzzer produces a 1200 Hz tone;
- the servo moves to 90°.

When the barrier closes, the red LED turns on, the green LED turns off, the servo returns to 0°, and a lower 600 Hz tone is produced.

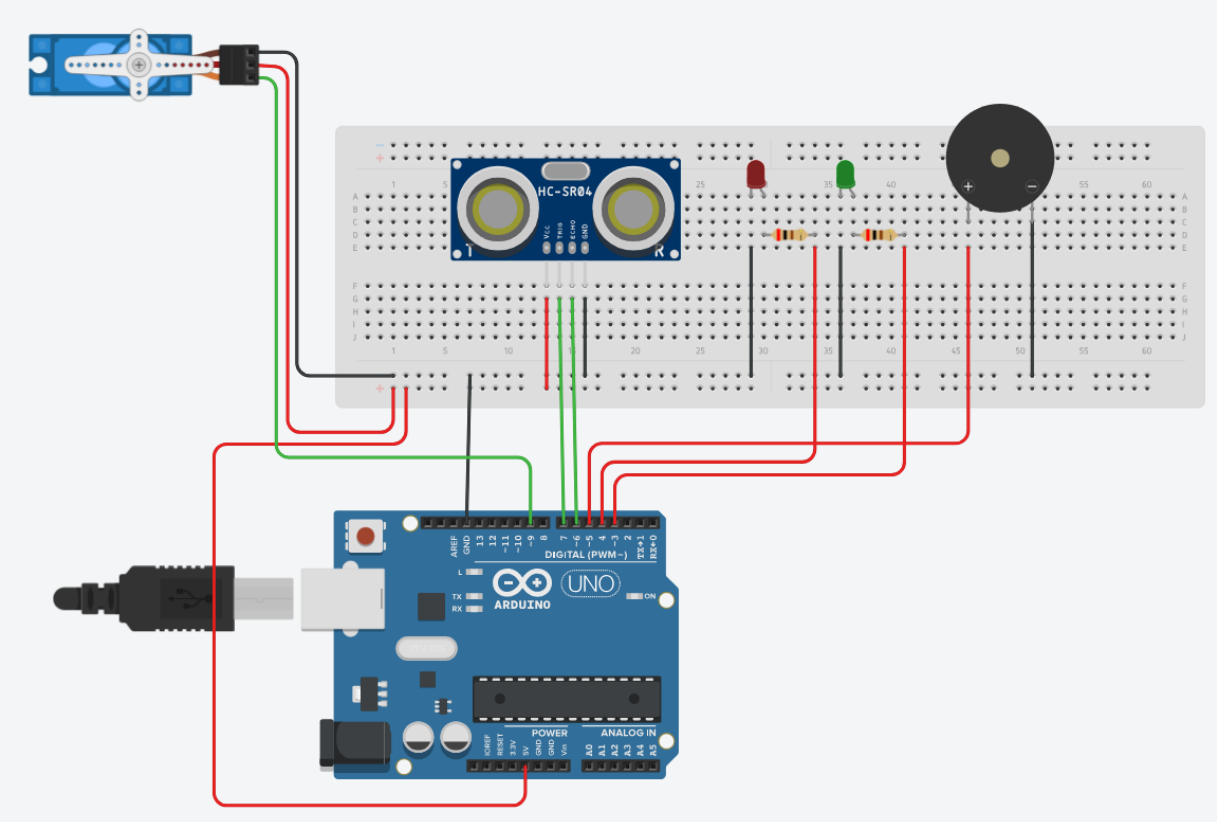


Figure 1: Complete automatic barrier prototype in Tinkercad.

3 Components and Connections

Component	Purpose	Arduino connection
Arduino Uno	Main controller	–
HC-SR04	Measures object distance	TRIG D7, ECHO D6
Micro servo	Opens/closes barrier	Signal D9
Red LED	Barrier closed indicator	D4
Green LED	Barrier open indicator	D3
Piezo buzzer	Audible movement indication	D5
$2 \times 200\ \Omega$ resistors	LED current limiting	LEDs
Breadboard	Component interconnection	5 V and GND rails

Table 1: Main components and Arduino connections.

4 Circuit Schematic

Figure 2 shows the electrical connections between the Arduino and the individual components. The Arduino 5 V and GND connections are distributed through the breadboard power rails so that the ultrasonic sensor and servo can share the same supply.

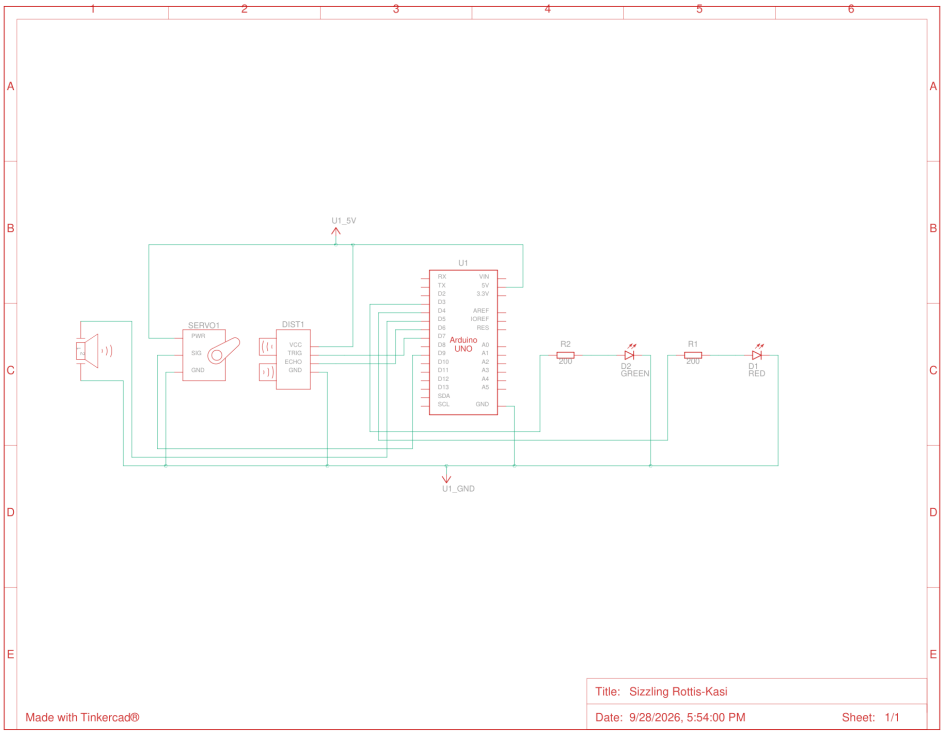
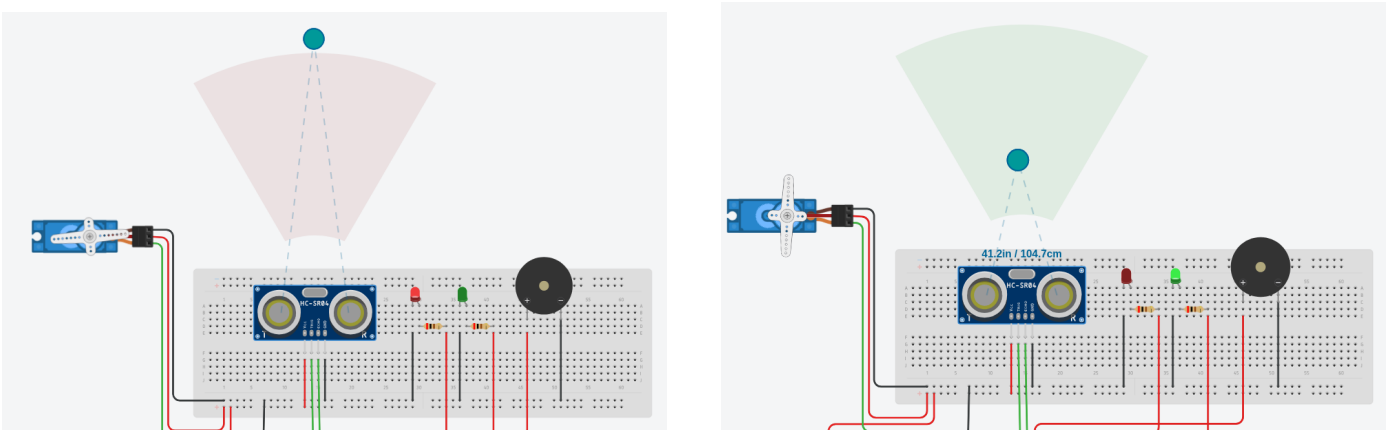


Figure 2: Tinkercad schematic of the automatic barrier system.

5 Testing and Results

The completed prototype successfully detects objects, changes the barrier position, controls the status LEDs, and generates different buzzer tones for opening and closing. Serial output is also used to display measured distance and the current barrier action.



(a) Barrier closed: red LED active.

(b) Barrier open: green LED active.

Figure 3: Prototype operation in the two main states.

The main limitation observed during testing was the servo movement speed in the Tinkercad simulation. The standard Servo library specifies the requested angle but does not directly control the physical movement speed. The ultrasonic sensor can also occasionally return no echo; this case is handled by the program and causes an open barrier to close after the configured timeout.

6 Possible Improvements

A physical version could be improved by adding a second sensor to determine whether a vehicle has completely passed the barrier before closing it. A display could show the measured distance or barrier state. Further improvements could include adjustable distance thresholds, a manual override, or a second sensor for additional safety.